Analysis I

Uebungsblatt 9

Problem:	Comment:	Points:
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We prove the equivalence of the three statements by showing the cycle of implications (a) $\implies$ (b), (b) $\implies$ (c), and (c) $\implies$ (a).

(a) $\implies$ (b): M is closed $\implies$ M contains all its accumulation points

Solution. We assume (a), that M is closed. By definition, this means its complement $M^c = X \setminus M$ is an open set. We must show (b), that M contains all its accumulation points.

We will use proof by contradiction. Let x be an accumulation point of M. Assume, for the sake of contradiction, that $x \notin M$. This means $x \in M^c$.

Since M^c is open, there exists a neighborhood $U \in \mathcal{U}(x)$ such that $U \subset M^c$. This means that U contains no points of M, so $U \cap M = \emptyset$.

However, x is an accumulation point of M. By Definition 4.2.10, this means that for *every* neighborhood $U \in \mathcal{U}(x)$, we must have $M \cap (U \setminus \{x\}) \neq \emptyset$.

This is a contradiction. Therefore, our initial assumption ($x \notin M$) must be false. This proves that $x \in M$, so M contains all its accumulation points. $\square$

(b) $\implies$ (c): M contains its accumulation points $\implies$ M is sequentially closed

Solution. We assume (b), that M contains all its accumulation points. We must show (c), that for any convergent sequence $(x_n)_{n\in\mathbb{N}}$ with $x_n \in M$ for all n, its limit $L = \lim_{n\to\infty} x_n$ is also in M.

Let (x_n) be such a sequence in M, converging to L. We must show $L \in M$. We consider two cases:

1. **Case 1: L is an accumulation point of M.** By our assumption (b), M contains all its accumulation points. Therefore, $L \in M$.

2. **Case 2: L is *not* an accumulation point of M.** This means there exists some neighborhood $U \in \mathcal{U}(L)$ such that $M \cap (U \setminus \{L\}) = \emptyset$. In other words, the only point of M that U is allowed to contain is L itself.

   However, we know that $\lim_{n\to\infty} x_n = L$. By the definition of a limit, for this specific neighborhood U, there must exist an $N \in \mathbb{N}$ such that for all $n \geq N$, $x_n \in U$.

   Since $x_n \in M$ for all n, this means that for all $n \geq N$, $x_n \in U \cap M$.

   From our case assumption, the only point in $U \cap M$ is L. Therefore, for all $n \geq N$, we must have $x_n = L$. But $x_N \in M$, which means $L \in M$.

In both possible cases, we conclude that $L \in M$. $\square$

(c) $\implies$ (a): M is sequentially closed $\implies$ M is closed

Solution. We assume (c), that M contains the limits of all its convergent sequences. We must show (a), that M is closed. We will do this by proving that the complement, M^c, is open.

To show M^c is open, we must show that for any $x \in M^c$, there exists a neighborhood $U \in \mathcal{U}(x)$ such that $U \subset M^c$.

Let's use proof by contradiction. Assume M^c is *not* open. This means there exists at least one point $x \in M^c$ such that for *every* neighborhood $U \in \mathcal{U}(x)$, U is not fully contained in M^c. This implies that for every $U \in \mathcal{U}(x)$, $U \cap M \neq \emptyset$.

Let's use this property to construct a sequence. For each integer $n \geq 1$, consider the open ball $\mathrm{B}(x, 1/n)$. This is a neighborhood of x. According to our assumption, this ball must contain a point from M. Let's pick one and call it x_n. So, for each $n \geq 1$, we have a point x_n such that $x_n \in M$ and $x_n \in \mathrm{B}(x, 1/n)$.

Now, consider the sequence $(x_n)_{n\in\mathbb{N}}$ we have just constructed:

- $x_n \in M$ for all n (by construction).
- The distance $||x_n - x|| < 1/n$ for all n.
- As $n \to \infty$, $1/n \to 0$. By the Squeeze Theorem, $\lim_{n\to\infty} ||x_n - x|| = 0$, which means $\lim_{n\to\infty} x_n = x$.

We have a sequence (x_n) entirely in M that converges to the limit x. By our hypothesis (c), this limit x must also be in M.

But this is a contradiction. We started by choosing x from M^c (so $x \notin M$). We have now proven $x \in M$. A point cannot be in both M and M^c. Therefore, our assumption that M^c is not open must be false.

This proves that M^c is open, and therefore M is closed.

Since we have shown (a) $\implies$ (b), (b) $\implies$ (c), and (c) $\implies$ (a), all three statements are equivalent. □

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Solution. (a) The statement $(\overline{M})^\circ = M^\circ$ is false in general.

Example: $X = \mathbb{R}$, $M := (0,1) \cup (1,2)$. Then

$$M^\circ = M = (0,1) \cup (1,2), \qquad \overline{M} = [0,2], \qquad (\overline{M})^\circ = (0,2).$$

Hence $(\overline{M})^\circ \neq M^\circ$.

(b) The statement $\overline{M^\circ} = \overline{M}$ is false in general.

Example: $X = \mathbb{R}$, $M := \{0\} \cup (1,2)$. Then

$$M^\circ = (1,2), \qquad \overline{M^\circ} = [1,2], \qquad \overline{M} = \{0\} \cup [1,2].$$

Hence $\overline{M^\circ} \neq \overline{M}$.

(c) The statement $(\overline{M})^\circ = \overline{M^\circ}$ is false in general.

Example: $X = \mathbb{R}$, $M := (0,1) \cup (1,2)$. As in (a),

$$\overline{M} = [0,2], \quad (\overline{M})^\circ = (0,2), \quad M^\circ = (0,1) \cup (1,2), \quad \overline{M^\circ} = [0,1] \cup [1,2].$$

Thus $(\overline{M})^\circ = (0,2) \neq [0,1] \cup [1,2] = \overline{M^\circ}$. □

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(a)

Solution. Let $z_0 \in \mathbb{C} \setminus \{0\}$. For $z \in \mathbb{C} \setminus \{0\}$ we compute

$$|f(z) - f(z_0)| = \left|\frac{1}{z} - \frac{1}{z_0}\right| = \left|\frac{z_0 - z}{z z_0}\right| = \frac{|z - z_0|}{|z|\,|z_0|}.$$

Choose $\delta_0 := \frac{|z_0|}{2} > 0$ and assume $0 < |z - z_0| < \delta_0$. Then

$$|z| \geq |z_0| - |z - z_0| > |z_0| - \frac{|z_0|}{2} = \frac{|z_0|}{2},$$

so

$$|f(z) - f(z_0)| \leq \frac{|z - z_0|}{(\frac{|z_0|}{2})\,|z_0|} = \frac{2}{|z_0|^2}\,|z - z_0|.$$

Given $\varepsilon > 0$, set

$$\delta := \min\left\{\delta_0,\ \frac{|z_0|^2}{2}\varepsilon\right\}.$$

Then $0 < |z - z_0| < \delta$ implies

$$|f(z) - f(z_0)| \leq \frac{2}{|z_0|^2}\,|z - z_0| < \frac{2}{|z_0|^2} \cdot \frac{|z_0|^2}{2}\varepsilon = \varepsilon.$$

Hence f is continuous at every $z_0 \in \mathbb{C} \setminus \{0\}$. Therefore f is continuous on $\mathbb{C} \setminus \{0\}$. □

(b)

Solution. On each open interval $(n, n+1)$ with $n \in \mathbb{Z}$, we have $\lfloor x \rfloor = n$ constant, hence

$$g(x) = nx^2, \qquad x \in (n, n+1),$$

which is a polynomial and therefore continuous on $(n, n+1)$. Possible discontinuities can only occur at the integers.

Fix $n \in \mathbb{Z}$ and consider $x \to n$.

For $x < n$ close to n we have $\lfloor x \rfloor = n - 1$, so

$$\lim_{x \to n^-} g(x) = \lim_{x \to n^-} (n-1)x^2 = (n-1)n^2.$$

For $x > n$ close to n we have $\lfloor x \rfloor = n$, so

$$\lim_{x \to n^+} g(x) = \lim_{x \to n^+} nx^2 = n^3.$$

The function value at n is

$$g(n) = \lfloor n \rfloor n^2 = n^3.$$

Thus g is right–continuous at every integer (right limit equals $g(n)$), and continuous from the left exactly when

$$(n-1)n^2 = n^3 \iff n^2(n-1-n) = 0 \iff n = 0.$$

- g is continuous on each open interval $(n, n+1)$ for all $n \in \mathbb{Z}$.
- g is continuous at 0, since $\lim_{x \to 0^-} g(x) = \lim_{x \to 0^+} g(x) = g(0) = 0$.
- For every integer $n \neq 0$ the left and right limits differ, so g has a jump discontinuity at each such n.

Therefore g is continuous on $\mathbb{R} \setminus (\mathbb{Z} \setminus \{0\})$, i.e. continuous at all $x \notin \mathbb{Z}$ and at $x = 0$ $\square$

(c)(i)

Solution. For $(x, y) \neq (0, 0)$ the function h is a quotient of continuous functions with nonzero denominator, hence continuous there. It remains to check continuity at $(0, 0)$.

Note that by definition $h(0, 0) = 0$.

For $(x, y) \neq (0, 0)$,

$$|h(x, y)| = \frac{x^2 + y^2}{|x| + |y|} = \frac{\|(x, y)\|_2^2}{|x| + |y|} \leq \frac{\|(x, y)\|_2^2}{\|(x, y)\|_2} = \|(x, y)\|_2,$$

because $|x| + |y| \geq \sqrt{x^2 + y^2} = \|(x, y)\|_2$.

Therefore, for all $(x, y) \neq (0, 0)$,

$$|h(x, y) - h(0, 0)| = |h(x, y)| \leq \|(x, y)\|_2.$$

Moreover,

$$\lim_{(x,y) \to (0,0)} \|(x, y)\|_2 = 0.$$

By the squeeze theorem, it follows that

$$\lim_{(x,y) \to (0,0)} h(x, y) = 0 = h(0, 0).$$

Thus h is continuous at $(0, 0)$ and therefore continuous on $\mathbb{R}^2$ with respect to the Euclidean norm. $\square$

(c)(ii)

Now let $\|\cdot\|$ be any norm on $\mathbb{R}^2$. We know that all norms on $\mathbb{R}^2$ are equivalent, so there exist constants $c, C > 0$ such that

$$c\,\|(x,y)\|_2 \le \|(x,y)\| \le C\,\|(x,y)\|_2 \quad \text{for all } (x,y) \in \mathbb{R}^2.$$

In particular, from $c\,\|(x,y)\|_2 \le \|(x,y)\|$ we obtain

$$\|(x,y)\|_2 \le \frac{1}{c}\,\|(x,y)\|.$$

From part (i) we already know that

$$|h(x,y)| \le \|(x,y)\|_2 \quad \text{for all } (x,y) \in \mathbb{R}^2.$$

Combining the two inequalities gives

$$|h(x,y)| \;\le\; \|(x,y)\|_2 \;\le\; \frac{1}{c}\,\|(x,y)\|.$$

Let $\varepsilon > 0$ and set $\delta := c\varepsilon$. Then, whenever $\|(x,y)\| < \delta$, we have

$$|h(x,y)| \;\le\; \frac{1}{c}\,\|(x,y)\| \;<\; \frac{1}{c}\,\delta \;=\; \varepsilon.$$

This shows that for the norm $\|\cdot\|$,

$$\|(x,y)\| \to 0 \quad \Longrightarrow \quad h(x,y) \to 0,$$

so h is continuous at $(0,0)$ with respect to the arbitrary norm $\|\cdot\|$.

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(a)

Solution. We show that the function

$$f(z) = \frac{z}{|z|}$$

cannot be continuously extended at 0.

Consider the sequences

$$z_n = \frac{1}{n}, \qquad w_n = -\frac{1}{n}.$$

Both satisfy

$$z_n \to 0, \qquad w_n \to 0 \quad (n \to \infty).$$

Now compute:

$$f(z_n) = \frac{z_n}{|z_n|} = 1 \quad \text{for all } n,$$

so

$$f(z_n) \to 1.$$

Similarly,

$$f(w_n) = \frac{w_n}{|w_n|} = -1 \quad \text{for all } n,$$

so

$$f(w_n) \to -1.$$

Since the limits along two sequences approaching 0 are different, the limit

$$\lim_{z \to 0} \frac{z}{|z|}$$

does not exist. Therefore, f cannot be continuously extended at 0. $\square$

(b)

We require continuity at the junction points $x = -1$ and $x = 1$.

At $x = -1$

$$\lim_{x \to -1^-} (x^2 - \alpha x + \beta) = 1 + \alpha + \beta, \quad g(-1) = \gamma, \quad \lim_{x \to -1^+} (\alpha + \beta)x = -\alpha - \beta.$$

Continuity at -1 gives

$$1 + \alpha + \beta = \gamma, \qquad -\alpha - \beta = \gamma$$

hence

$$1 + \alpha + \beta = -\alpha - \beta \;\Rightarrow\; 2\alpha + 2\beta = -1. \tag{I}$$

At $x = 1$

$$\lim_{x \to 1^-} (\alpha + \beta)x = \alpha + \beta, \quad g(1) = \delta, \quad \lim_{x \to 1^+} (x^2 + \alpha x - \beta) = 1 + \alpha - \beta.$$

Continuity at 1 gives

$$\alpha + \beta = \delta, \qquad 1 + \alpha - \beta = \delta$$

hence

$$\alpha + \beta = 1 + \alpha - \beta \;\Rightarrow\; 2\beta = 1 \;\Rightarrow\; \beta = \tfrac{1}{2}. \tag{II}$$

Solving

Insert (II) into (I):

$$2\alpha + 2 \cdot \tfrac{1}{2} = -1 \;\to\; 2\alpha + 1 = -1 \;\to\; \alpha = -1.$$

Then

$$\gamma = -\alpha - \beta = 1 - \tfrac{1}{2} = \tfrac{1}{2}, \qquad \delta = \alpha + \beta = -1 + \tfrac{1}{2} = -\tfrac{1}{2}.$$

Thus

$$\alpha = -1, \quad \beta = \tfrac{1}{2}, \quad \gamma = \tfrac{1}{2}, \quad \delta = -\tfrac{1}{2}.$$

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